Program: **Diploma in Communication and Computer Networking**	
Course Code: **3322**	Course Title: **Logic Design**
Semester: **3**	Credits: **3**
Course Category: **Programme core course**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- Design and implement combinational logic circuits using Logic Gates.
- Design and implement sequential logic circuits using Integrated Circuits.

Course Prerequisites: NIL

Course Outcomes:

On completion of the course, the student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Understand digital systems and data representation & logic gates.	12	Understanding
CO2	Understand combinational logic gates.	11	Applying
CO3	Design sequential logic circuits.	10	Applying
CO4	Analyze D/A and A/D and memory systems.	10	Applying
	Series Test	2	

CO–PO Mapping

Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**
CO1	3	2	1		2		
CO2	3	1	2	1	2		
CO3	3	2	1	2			
CO4	3	2	1		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand Digital Systems and Data Representation &Logic Gates**		
M1.01	To Understand Digital Systems.	3	Understanding
M1.02	To explain various Number Systems.	2	Understanding
M1.03	To Describe Logic gates.	3	Understanding
M1.04	To explain Boolean Algebra.	4	Applying
Contents: **Digital Systems & Logic Gates , Combinational Logic** Digital Systems – Binary numbers – Number base conversions- Octal, Hexadecimal - Complements of Numbers – Signed Binary Numbers. Boolean Algebra -Basic Theorems and Properties of Boolean Algebra – Boolean Functions-Canonical and standard forms — Digital Logic Gates.			
CO2	**Acquire knowledge about basic memory organization.**		
M2.01	To understand K-Map.	2	Understanding
M2.02	To describe SOP and POS minimization.	4	Applying
M2.03	To explain different Combinational Circuits.	4	Applying
	Series Test –I	1	
Contents: **Combinational Logic Gates** The K- Map Method -Product –of-Sums & Sum-of-Products Simplification – Don't Care Conditions – NAND and NOR Implementation –Combinational Circuits – Binary Adder – Subtractor.			
CO3	**Develop knowledge about processor structure and functions.**		
M3.01	To describe Sequential Circuits.	3	Understanding
M3.02	To explain Storage elements – Latches &Flip-Flops.	3	Understanding

M3.03	To explain different registers and counters.	4	Applying
Contents: **Sequential Logic:** Sequential Circuits :Storage elements – Latches & Flip-Flops - Ripple Counter - Synchronous Counter-Ring counter - Johnson Counter.			
CO4	**Implement the Control Unit.**		
M4.01	Study basic concept of DAC & ADC.	4	Understanding
M4.02	To describe Memory systems.	5	Applying
	Series Test–II	1	
Contents: DAC –ADC, Random Access Memory -Memory decoding- Read Only Memory			

Text/Reference

T/R	Book Title/Author
T1	M. Morris Mano, Digital Logic & Computer Design, 4/e, Pearson Education, 2013
T2	Thomas L Floyd, Digital Fundamentals, 10/e, Pearson Education, 2009.
T3	M. Morris Mano, Computer System Architecture, 3/e, Pearson Education, 2007.
R1	M. Morris Mano, Michael D Ciletti , Digital Design With An Introduction to the Verilog HDL, 5/e, Pearson Education, 2013.
R2	Donald D Givone, Digital Principles and Design, Tata McGraw Hill, 2003
R3	Kandel Langholz, Digital Logic Design, Prentice Hall, 1988.
R4	Rafiquzzaman & Chandra, Modern Computer Architecture, West Pub. Comp., 1988.
R5	Basavaraj,B., Digital fundamentals, New Delhi: Vikas Publishing House, 1999

Online Resources

Sl No	Website Link
1	**http://nptel.ac.in/course**:s/Webcourse-contents/IIT-%20Guwahati/comp_org_arc/web/
2	https://www.tutorialspoint.com/computer_logical_organization/combinational_circuits.htm

3	https://www.electronics-tutorials.ws/sequential/seq_6.html
4	https://www.tutorialspoint.com/digital_circuits/digital_circuits_boolean_algebra.htm